Overview, selecting and searching data Activity Book

Create Visualizations with Qlik Sense

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Activity 1: Exploring the Hub and analyzing an app

Summary:

- Gain familiarity with the Qlik hub.
- Open an app, identify the parts or views in the app, and navigate through them.
- Make simple selections and searches, apply bookmarks, and play stories.

Qlik Sense app: ABC Sick Leave

Challenge

- Familiarize yourself with the Qlik Sense hub, check the options to understand the platform you are using.
- Import or copy and move the necessary Qlik Sense apps for this module into your Personal space.
- Navigate through the apps and review the information related to each app.
- Search for the **Sick Leave** app, open it, and identify components in the app (sheets, bookmarks, stories).
- Analyze the data by making selections and be prepared to explain the following using existing visualizations and bookmarks:
  - Sick leave ratio for male - female groups in different regions and years (Example: *China 2022*, *China 2023*, etc.)
  - Correlation between sick leave days and overtime hours
- Explore other selections and searches:
  - Compare Sick leave and Overtime hours for 2023 vs. China 2023 vs. USA 2023 using the Selection states sheets.
  - Find all the information about an employee (Ricardo Gucci) in the Sick Leave sheet using the Global search.

Qlik Sense Client-Managed: If you are working with the Skytap environment, the app or apps will be available in the stream for this module. Right-click on the app to duplicate them to move them to your Work area or personal area.

Qlik Sense Desktop: Place the .qvf file on the path **Documents\Qlik\Sense\Apps**. Then the app will appear on the Qlik Sense Desktop hub.

Hints

- Visualizations in an app can be used to make selections. You can select one or more bars in the bar chart to reduce the data and make selections.

Step-by-step guide

1. Discuss which Qlik Sense platform are you going to use for this activity.
2. Navigate to the **Qlik Sense hub** and explore the options and navigation menus. Find the **Help** link as one of the options under the navigation menu in the top right area. Go to the **Resource Center** (between the **Add new** button and **Notifications** icon) and check out the suggested resources depending on your user type.

   Qlik Sense Client-Managed: When selecting the **About** option you can see the version of the product you are using. In Qlik Sense Cloud editions, you are working with the latest version available.

3. Import or duplicate the apps into your Personal area, if needed:
   a. To import the apps, select **Add new > Upload app** option.
   b. To duplicate apps, click ••• on the app and select **Duplicate**. Then move them to your personal area.

   Follow your instructor in this process to populate your Hub with the apps.

   Qlik Sense Desktop: Locate the app or apps on the local folder **Documents\Qlik\Sense\Apps**.

   Qlik Sense on Windows / Qlik Sense Desktop: Right click over the app icon to duplicate it.

4. In the hub, hover over each app and see the related information. Then search for *Sick Leave* to filter the available apps.
5. Open the **ABC Sick Leave** app. Explore in the **App overview** to see how many sheets, bookmarks, and stories are in this app. You can also check other views and areas in the app, like the **Data model viewer**, under the **Prepare** section, to see the tables, fields, and data associations.
6. Navigate to the **Sick Leave** sheet, and analyze the visualizations. What is the *Male-Female* ratio of sick days?
7. Make a selection by clicking on *China* in the bar chart. Confirm your selection by clicking on the green check mark icon. All the visualizations update with this selection because the data is associated, and the selection appears in the selections bar.

   The pie chart shows the male-female ratio of sick leave for *China*. You can change the dimension using the drop down arrow next to *Gender*.

8. Click the bookmarks **>** button and select China 2022. Notice how the current selection changed. Review the pie chart values again.
9. Using the selection bar at the top change the selected year to 2023 and click the**>** bookmarks button, then select **Create new bookmark**, to save the current selection (China and 2023) as a new personal bookmark.
10. Click **Clear all selections** **:** and review the charts (combo chart and scatter plot) at the bottom of the sheet that show a correlation between sick leave days and overtime hours.
11. Navigate to **Storytelling** and play the story called *Sick Days and Overtime analysis*.
12. Go back to the **Analyze** view and navigate to the sheet called *Selection states*. In this sheet we identify three sets (Default state, State1 and State2). Each set of charts reflect only the

selections in the corresponding state they are in. The bar chart on the right combines all the three states to compare them in the same chart. This allows you make comparisons easily.

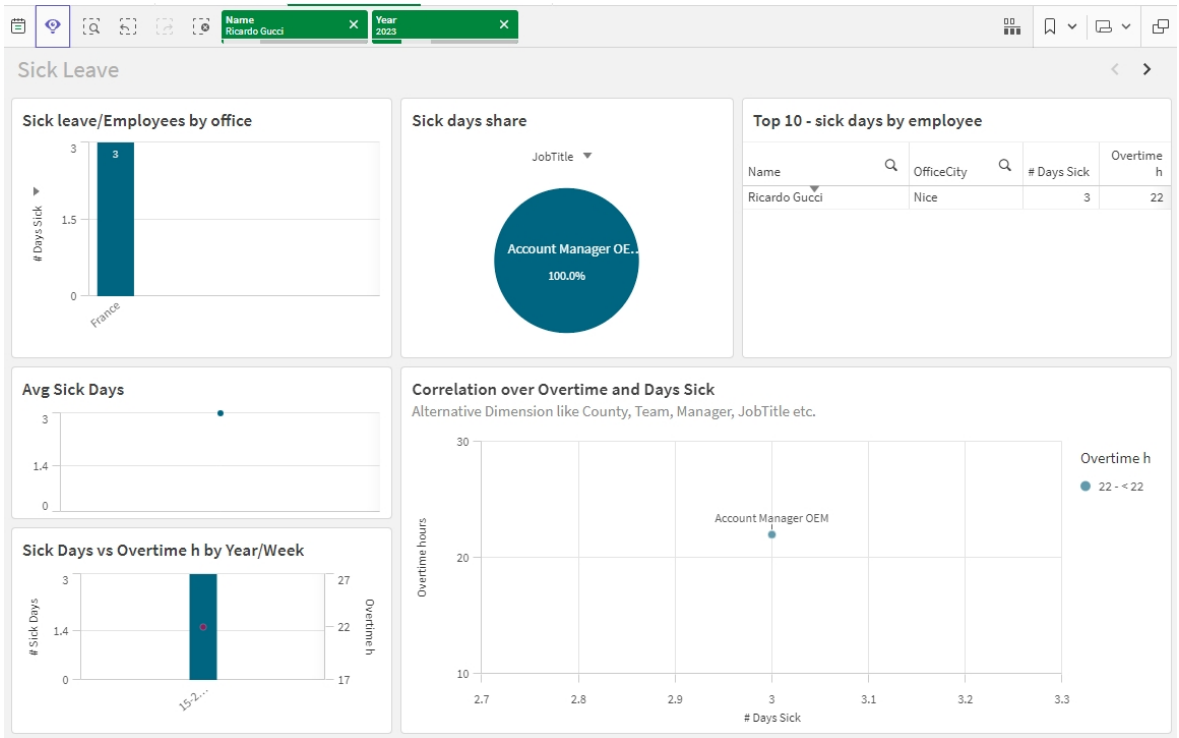
Select the following:

a. Year 2023 in the Default state
b. Year 2023 and China in the State1
c. Year 2023 and USA in the State2

Notice how the different selections are reflected in the selection bar at the top.

13. Navigate to the previous sheet, clear all selections, and use the Smart search (Global search) icon at the top to type *Gucci* in the search box. Select *Ricardo Gucci*. Type 2023 and select that value on Year. Review all the info available in the sheet: Ricardo Gucci is a male employee that works for the Nice office (France) as Account Manager OEM in the Sales team. In 2023 he had 3 sick days and worked 22 overtime hours.

Solution



Activity 2: Analyzing dimensions and measures

Summary:

- Identify different visualizations and the dimensions and measures in them.
- Change visualizations using the alternative dimensions and measures and the Exploration menu.

Qlik Sense app: Vintage movies

Challenge

Open the **Vintage Movies** app in a managed space or stream in the hub, navigate through the sheets, and analyze the visualizations.

Qlik Sense Desktop: Qlik Sense Desktop does not offer streams or managed/shared spaces. Locate the app in your personal space to work in this activity.

Qlik Sense Business: Managed spaces are not supported in this version but you can use shared spaces or your personal space for this activity.

What are the main dimensions and measures, and what visualizations are being used in this sheet?:

- **Dashboard** sheet
- **Dimensions and Expressions (1)** sheet

Explore how you can change the visualizations without editing the sheet:

- In the **Movies and Director Analysis** sheet, adjust the alternative dimensions and measures in the charts.
- In the **Movies across time** sheet, use the Exploration menu to change how the bar chart looks.

Hints

- A dimension groups the data in the visualization and a measure is usually aggregated with a function.
- When a visualization has reduced space, some of the options might not be visible. Hover over the chart and click **S** in the top right corner to work with the object in full screen mode; click **E** to return to normal view.
- You can interact with select properties in some visualizations while in analysis mode. When the app is opened in analysis mode (not edit), use the **Exploration menu Ú** by right-clicking over the chart.

Step-by-step guide

1. Open the **Vintage Movies** app, then, from the **App overview**, navigate to the **Dashboard** sheet.
2. Review the charts and visualizations in this sheet:
    - At the top we find three KPIs representing main measures for the analysis. The first KPI includes a secondary measure, too.

- Three bar charts displayed with a horizontal layout show the distribution of the measures across some dimensions (decade, director, actor). Notice that when selecting a decade in the first bar chart, the dimension changes and drills into the years of that selected decade.
- Additional information about the trend of the number of movies (line chart) and the share of the rating (donut or pie chart) can be found in the sheet.

3. Navigate to the **Dimensions and Expressions (1)** sheet. Review the different type of charts and identify what measures and dimensions are being used in them.

Solution

- The first visualization is a combo chart that represents the number of movies and average ratings per decade:
  - Dimension: Decade
  - Measures: Number of movies (count) and Avg rating (average)
- The second visualization is a gauge and represents the average rating over all the movies. There is no dimension in this chart.
  - Measure: Avg rating (average)
- The third visualization is a pie chart that displays the share of movies between the ratings.
  - Dimension: Rating
  - Measure: Number of movies (count)
- The fourth visualization is a line chart displaying a trend of the number of movies over time (by year) where each line represents a rating.
  - Dimensions: Year and Rating
  - Measure: Number of movies (count)

4. Navigate to the **Movies and Director Analysis** sheet and interact with some of the charts. Note that you can expand them hovering over them and viewing them in **Full screen** mode (**S**).
5. Use the *Number of films and average rating over time* chart to select one decade and see the yearly trend number of movies and ratings. In this case, a drill-down dimension is being used to display the hierarchy between decades and years in the decades. Also, in this chart, there are two Y-axes scales (left and right) as the two measures used in the chart have different scales.
6. Use the drop down arrows next to the measures and dimension names in the axes of the visualizations to change the measures and dimensions picking other alternative ones that were preconfigured by the app author.
7. Navigate to the previous sheet, **Movies across time**, and clear selections. Notice how the text at the top of the sheet is dynamic text and it calculates some expressions based on the current selection.
8. Hover over the bar chart, select the **More** menu and open the **Exploration menu**. The chart expands to full screen and the **Exploration menu** becomes visible.
9. In the Exploration menu, change the visualization properties:
   - In the **Presentation** section, change the orientation of the bars and the display from stack to grouped.

- In the **Sorting** section, sort by *Decade-Year* ascending, then *Rating* ascending, and after them, *Number of movies* ascending.
- In the **Data** section, select the alternative measure.

10. You need to click **Apply** to save the changes. Note that you can use the **Discard** button to return the chart to the original state.

Solution

Activity 3: Answer data questions by making selections in an app

Summary:

- Understand the data and find answers to data questions in a Qlik Sense app.
- Appreciate the Qlik associative engine and learn how to make selections in different areas of the app.
- Explore the selection states, the Selections menu, and the Selections bar.
- Work with the Selections tool, learn how to make advanced selections, and review Associative insights.

Qlik Sense app: Vintage movies

Challenge

Use the **Vintage Movies** app to answer the following questions:

- **Question 1**: Within the *1950's*, how many movies are there with a rating of *4*? How many of them have a length up to *120* minutes?
- **Question 2**: How many movies were created between the years *1970* and *1980* with a rating of *4* or *5*? Who were the most productive directors and actors in that period?
- **Question 3**: During which decades did *John Ford* direct his movies? How many movies did *John Ford* direct in the *1950's* and *1960's*?
- **Question 4**: How many movies from the *1950's* were not directed by *John Ford?* Review how many directors were involved on those movies and find out other details about them (using Associative insights).
- ***Question 5**: Did John Ford and Ava Gardner work together in the 1950's? If yes, which movies?*

Hints

- Use the different sheets in the app to get the required information for the questions.
- Filter panes, chart axes labels, and table columns can be used to make selections.
- The **Selection toolbar** and the **Selection menu** allow you to make further selections and review current selections.
- The Selections tool allows you to make selections in all the fields in the app. From this tool you can access the **Associative insights** panel to generate insights about the selections and the non-selected values based on the selection.

Step-by-step guide

1. Open the **Vintage Movies** app and navigate to the **Movie selector** sheet in analysis mode.

Question 1

2. Select *1950's* in the *Decade* filter pane. Then, in the table, select the value *4* in the *Rating* column and confirm the selection. Look at the dynamic text on the *Titles* filter pane. There are 52 titles from the 1950's decade with a rating of 4.

3. Using the **Selections tool** icon ( ) at the top right, click the magnifying glass icon for the *Length* field and type *<=120*, then press enter to select the values associated in the search.

   Confirm the selection and click **Selections tool** ( ) again to close the Selections tool. From those movies in question 1, there are 38 movies that are less than or equal to 2 hours long.

Question 2

4. Navigate to the **Dashboard** sheet and clear all selections. Select the segments in the donut chart that correspond with the ratings *4* and *5*, and then perform a range selection in the line chart using the horizontal axis for the years *1970-1980*. There were 57 movies.

   > **Note**: If you don't see the ratings in the donut chart or the year values in the x-axis, increase the size of your window or view the chart in full screen mode.

5. Look at the bar charts to find out who was the most productive director and actor in those movies. *Woody Allen* and *Donald Sutherland* were the director and actor that were most productive during that period in those movies with ratings of *4* and *5*.

Question 3

6. Clear all selections and select *John Ford* using the *Number of movies by directors* bar chart in the middle of this sheet. Look at the values on the bar chart at the left. *John Ford* directed his movies from the 1920's to the 1960's.
7. Select the decades *1950's* and *1960's* using the bar chart on the left. *John Ford* directed 23 movies during those decades.

   > Note: The selected fields are displayed in the **Selections toolbar**. You can click on the *Decade* selections to see the selected values. The selected decades display in green and the rest of the decades appear in light gray and dark gray. Looking at the decades you have selected, explain why the remaining decades are displayed on a light gray or dark gray background.

Question 4

8. Using the Selection toolbar on the top, click the *Decade* filter, and deselect *1960's*, so the only remaining selected decade is *1950's*. Then, using the Selection toolbar, select the *Director* filter. You can see that *John Ford* is the only selection. Click the **Selection menu** ¥ and click **Select alternative** to select all the directors related to the selected decade excluding *John Ford*.
9. Look at the *Number of movies* KPI on the dashboard. There are 309 movies associated with the selection, but this number is not the correct answer. Movies can be directed by more than one director. In this case, the movie called *Mister Roberts* is directed by *John Ford* and *Mervyn LeRoy* and this movie is being included in the count. You can see the details in the **Movie Selector** sheet. In order to get the correct answer, you need to first exclude the movies directed by *John Ford*.

Movie Details

Title	Director	Year
Mister Roberts	John Ford	1955
Mister Roberts	Mervyn LeRoy	1955

10. In the **Dashboard** sheet, step back in the selections using the **<** button in the Selection toolbar. The selected values are *John Ford* and *1950's*. Then click the **Selections** icon to display the **Selections tool**.
11. Click on the *Title* field, and using the **Selection menu ¥**, click **Select possible** to select all the movies directed by *John Ford* in the *1950's*.
12. Clear the selections in the *Director* field. Then click on the *Title* field again, and using the **Selection menu¥**, click **Select excluded**.
13. Review the **Associative insights** panel on the right. If this panel is not visible, click **Explore selections** to display it. Then click **Generate insights**. The insights cards appear in the panel. From the drop down box, select the **Measure**: *Number of movies*. Then click on the first card to review the details: 308 movies.
14. Review the cards on the right, under the **Insights from excluded values** area. You can add new fields to review insights from the **Add a dimension** button at the bottom. Select the *Director* card and click to select that card from the Insights from excluded values section to see that in total there are 133 directors (21.8%) related with the selection that created a total of 843 movies across all decades.

Question 5

15. Click **Selections** to hide the **Selection tool** and return to the sheet. Step back in the selections using the **<** button in the Selection toolbar to see all the titles directed by *John Ford* in the *1950's* selected again.
16. Scroll down in the *Number of movies by actor* bar chart on the right until you can see *Ava Gardner*. Select the bar corresponding with *Ava Gardner*. You can see *John Ford* and *Ava Gardner* worked together in 1 movie. Now open the **Selections tool** to see the *Title* field. The movie is *Mogambo*.

Activity 4: Working with bookmarks

Summary:

- Explore and work with bookmarks in a Qlik Sense app.
- Understand how to apply bookmarks and change them.
- Set a default bookmark in the app.

Qlik Sense app: Vintage movies

Challenge

In the **Vintage movies** app, create a bookmark, *1990TopMovies*, that will take you to the **Movie Selector** sheet with the following selections:

- *Decade — 1990's*
- *Rating — 4* and *5*

Then create a second bookmark, *1990BadMovies*, with the same behavior but selecting *1990's* and *1*.

Hints

- Bookmarks can be created from a sheet in analysis mode and when applied, they take you to the sheet they were created from.
- You can use the search or the selection panel to make the selections required in the *Decade* and *Rating* fields.
- In order to test the bookmark's navigation, you can apply the bookmark from another sheet in the app.

Step-by-step guide

1. Open the **Movie Selector** sheet in the **Vintage movies** app, and clear all selections, if necessary.
2. Use the search box to find the following values and then select:
   a. *Decade—1990's*
   b. *Rating—4,5*
3. Using **Bookmarks** in the toolbar, click **Create new bookmark**, and give the bookmark a name: *1990TopMovies*.
4. Change the selections in the *Rating* field to select *1*.
5. Using **Bookmarks** in the toolbar, create a second bookmark called *1990BadMovies*.
6. Clear all selections and navigate to the **Dashboard** sheet.
7. Using **Bookmarks** in the toolbar, select the *1990TopMovies* bookmark. The **Movie selector** sheet opens with the predefined selections in *Decade*, and *Rating*.
8. Using the **Bookmarks** button again, **right-click** on the *1990TopMovies* bookmark and select **Set as default bookmark**. This will set the bookmark as default for the app. The selections of this bookmark will be applied to the app when the user opens the app.

Activity 5: Searching for data and generating insights

Summary:

- Gain familiarity with the different searching options in a Qlik Sense app.
- Make searches in filter panes (fields) and use the Smart search (global search) to find answers to data questions.
- Learn how to generate insights and search with natural language queries in the app.
- Use the predefined Analysis Types available on the Insight Advisor.

Qlik Sense app: Vintage movies

Challenge

Using the **Vintage Movies** app, perform searches and use the Insight Advisor to answer these questions:

- **Question 1**: What movies were directed by *Francis Ford Coppola* in the *1980's*? Search for the arguments one by one — first *director*, then *decade*. Try to enter the search terms all at once. Do you get the same results as before?
- **Question 2**: What movies were directed by *Alfred Hitchcock*? Can you find movies directed by *Alfred Hitchcock* in the *1980's*?
- **Question 3**: What movies starred Nicolas Cage **or** Sofia Coppola? And which of those movies star both (Nicolas Cage **and** Sofia Coppola)?
- **Question 4**: What movies contain "*Godfather*" in the title or description? From those movies, who directed the ones that do not include the title,"*The Complete Epic*", their length is under *3* hours (*180* minutes), and *John Wayne* is listed in the movie's cast?
- **Question 5**: How many movies (*Titles*) are directed by more than one director in the 1980's and 1990's? Create a chart with the number of directors per movie for all decades using the Insight Advisor with a search.
- **Question 6**: Which directors worked with the highest number of actors? How is the behavior of the average rating along the years, depending on the length range? For answering these questions, create two charts using the Insight Advisor analysis types.

Hints

- The Smart search (Global search) can be launched from the Smart search icon on the top left corner or by using the keystroke shortcut Ctrl+F.
- You can also perform searches in filter panes or in the fields available in the Selections tool or in the Selection bar.
  - The **Search tool** allows you to search in values of all fields in the data model, and make selections on the results that match the search criteria.
  - In the **Selections tool** you can search in fields that are not represented on a sheet.
- The Insight Advisor (visual search) can also generate visual results for your searches and natural language queries. The **Insight Advisor** button can be found on the top bar on the left.
  - When using the Insight Advisor, you can use the left panel to choose the master dimensions and measures or fields you want to use in the search criteria, for example: "Title" by "Number of directors".

- When reviewing results in the Insight Advisor, you can navigate to the sheets where the visualizations were found or add new visualizations if the results did not exist in the app.
- Advanced searches with natural language can generate results with complex expressions and filters that can be saved in the app as master items for reusable purposes.
- The Analysis types are a more direct and guided way to create charts, with predefined templates.

Step-by-step guide

1. In the **Vintage Movies** app, navigate to the **Dashboard** sheet and ensure that all selections are cleared.

Question 1

2. Use the Smart search or Global search (**Ctrl+F**) and type *francis* in the search input box. The Smart search results show the *Title*, *Director*, and *Actor* fields. Select *Francis Ford* in the auto-suggestion list. The search results are filtered down to two possible results: *Francis Ford* the actor and *Francis Ford Coppola* the director. Choose the director in the search result.

> **Note**: The search is not case-sensitive. The **Smart search** displays a set of possible search strings in the auto-suggestion list when you type in the search box.

3. Search now for *1980's* and select the result under *Decade* in the results list. In the **Selections bar**, *Director* and *Decade* should be selected. Navigate to the **Movies Details** sheet to see all the movies related with the selection.
4. Clear all selections and start a new search for *francis 1980's*. The results of each term are highlighted in a different color. Select the row with *Director*: *Francis Ford Coppola* and *Decade*: *1980's*. The selections are applied.

Question 2

5. Search for *hitch* using the Smart search. There are no matches based on the current selections, but select the option **Clear selections and search again**. Then select *Alfred Hitchcock* under *Director* in the search results. Review the titles associated with this selection.
6. Clear all selections and search now for *1980's hitch*. In this case, there are no exact matches but select **Show partial matches**. From the terms matched list, select *Alfred Hitchcock* under *Director*. This provides a list from all decades.

Question 3

7. Clear all selections and navigate to the **Movie selector** sheet.
8. In the *Actor* filter pane, search for *Nicolas + Sofia* and select *Nicolas Cage* and *Sofia Coppola*. Take a look at the possible values in the *Title* filter pane. There are five movies listed and you can see the actors for each movie in the final column of the table at the bottom.
9. Deselect *Sofia Coppola* to only have *Nicolas Cage* selected in the *Actor* filter pane. Then, in the *Title* filter pane, select all the possible values (all the titles associated with *Nicolas Cage*).
10. Clear the selections in the Actor field. The movies are still selected because the selections in the *Title* field remain. Then, in the *Actor* filter pane, select *Sofia Coppola*.
11. Analyze the selections now in the *Title* filter pane. It shows the two movies in which both *Nicolas Cage* and *Sofia Coppola* appeared. Also notice that one of the three *Nicolas Cage* movies, *The Cotton Club*, now appears as selected excluded.

Question 4

12. Navigate to the **Dashboard** sheet and clear all selections. Then type *godfather* in the search tool. There's only one result matching the search, the *Title* field. There could be multiple values in the field that match the search criteria. Select the second result to make the selection in the *Title* field. This result includes the title containing *Godfathers* as well. You can see that there are 5 movies selected based on the search.
13. In the **Selections bar**, click the *Title* filter and clear the selection for *The Godfather: The Complete Epic* title. Look at the bar chart in the sheet to confirm the same two directors remain associated with the selected four movies.
14. Open the **Selections tool** and locate the *Length* filter. Then search in this filter, typing *<180*, and select all the values matching the search criteria. The movie length is displayed in minutes. This search finds all the possible movies with a duration under 3 hours.
15. In the *Actor* filter pane, use the search option to find *John* *. The list shows all actors with *John* as the first name. Then select *John Wayne* from the list, and the number of possible titles is reduced to only those where this actor has starred in. Look at the *Director* filter pane, the only director associated with those movies is *John Ford*.

Question 5

16. Clear all selections and select the **Insight Advisor**. You can type your question directly in the input box at the top or you can select the fields or master items on the left panel.
17. Type *Title by Number* in the input box. A suggested option list appears in the search box. From the suggested options, select *Number of Directors* and press enter to see the results. Complete the search text adding the following: *during the decade 1980's and 1990's* and press enter.
18. Review the results. A chart is the displayed as a **Matching result**. Notice other results are categorized into:
    a. **Additional results**, new generated charts based on the search. The **Add to sheet** button at the bottom will add it into a sheet in the app.
    b. **Existing charts**, links to existing visualizations in the app. The bottom right corner of each result includes the option to navigate to the sheet where the result can be found.

> **Note:** The generated **Insights** results are new visualizations generated based on your search and the data associations. You can add the new visualizations into your existing app by using the **Add to sheet** button. From the drop down menu next to this option, you can also choose the sheet the new visualization will be added to.

19. Select the matching result. It is a bar chart of the number of directors by movie titles for the specified decades. Note that you can perform selections in the chart. Additionally, in the right panel, you can see the dimensions, measures, and filters applied to the chart. Clear the filter to see the data for all decades.
20. Add the new chart to the **Dashboard** sheet. Navigate to the sheet using the **Go to sheet** icon in the bottom right corner of the chart result.
21. The new chart has been located on this sheet. You can use the Y-axis to select *Number of directors* from *1.1* to *12* to select all the movies directed by more than one director. Review in the selections bar that there are in total *52* movies directed by more than one director. You can now analyze what movies these are.

22. Right-click on the bar chart and select **View data**. The bar chart is transformed into a table. Notice that you can sort the table by the *Number of directors* by clicking on the column header. Then right-click on the table and select **View chart** to transform it back to a bar chart.

Question 6

23. Clear all selections and select the **Insight Advisor** again. This time, we will select a pre-defined analysis type from the tiles we have on the start page. Pick **Ranking** from the Analysis Types. Select *Number of Actors* from measures, and *Director* from dimensions.
24. Add the new chart to a new sheet. Go to that sheet and call it *Insights Analysis Types* sheet.
25. Come back to the Insight Advisor and pick the **Trend over time** analysis. Select *Avg rating* from measures, *Year* as Date/time dimensions, and *Length Range* from dimensions.
26. Add the new chart to the *Insights Analysis Types* sheet.

(Optional) Challenge 6: Analyzing the ABC Sales Dashboard app

Summary:

- Apply the concepts you learned in this module by answering business questions.
- Explore the data using the Selections tool and review the data relationships with the Qlik associative engine.
- Make selections and use the Qlik search capabilities to find answers in the data and generate useful insights.

Qlik Sense app:ABC Sales Dashboard FINAL

Challenge

Use the **ABC Sales Dashboard FINAL** app to answer the following questions about the data:

- **Question 1:** What is the sales share in the *EMEA* region and how many customers are in this region? Identify the top 3 customers in this region based on sales and the list of products never sold in the region.
- **Question 2:** What is the order volume range in *North America* and *South America*? Are there any outliers in these regions? If yes, provide the customer name and order volume.
- **Question 3:** Including *Bruxelles* and all the area 400 km around this city, how many customers are in that area, and what is the total sales generated in the last year in this area?
- **Question 4:** How many individual orders over 11.5k sales were made and by how many customers? Where are those customers located and what products did they buy in those transactions? For those customers, what is the *Margin%* on *Runner shoes* in this last year?
- **Question 5:** *Don Balon* is one of the VIP customers in the company. Identify what products were not sold in Don Balon's country. From those products, can you tell which ones were not sold to any other customers and what is their *UnitCost*?

Note: The objective of this **challenge** is to test whether you feel comfortable with the concepts that have been discussed so far in your class. There is no **Step-by-Step** section in this activity, so you must only rely on the **Hints** below to understand the tasks you have to execute.
As you are still new to many of these concepts, please don't hesitate to reach out for more guidance (help file, instructor, Qlik Community).

Hints

- Make selections in the different filter panes, charts, and tables in the sheets, or use the Selection tool to make specific selections.
- Notice how the Selection tool allows you to review associative insights to discover extra information about the data.
- Use the Smart search or the Insight Advisor to search for specific values or criteria in the app.